

Brain Tumor MRI Classification Using Machine Learning

1. Introduction

This project focuses on classifying brain MRI images into four categories using Machine Learning: Glioma, Meningioma, Pituitary, and No Tumor. Image-processing techniques are used to extract useful features, followed by a machine learning model for classification.

2. Dataset

The project uses the Brain Tumor MRI Dataset from Kaggle. It contains 5,600 training images and 1,600 testing images across four classes: Glioma, Meningioma, No Tumor, and Pituitary.

3. Methodology

Workflow: MRI Image → Preprocessing → Feature Extraction → Feature Selection → Extra Trees → Prediction

Preprocessing: Images are converted to grayscale, resized to 128×128 , Gaussian blurred, and normalized.

Feature Extraction: GLCM features (contrast, dissimilarity, homogeneity, energy, correlation), intensity features (mean, standard deviation, variance, minimum, maximum, median), and HOG features are used. Together they produce 1,775 features.

4. Feature Selection

Features	Accuracy
250	91.79%
500	92.59%
750	92.05%
1000	92.14%

The top 500 features were selected because they gave the highest accuracy among the tested feature counts.

5. Model

Extra Trees Classifier was used as the final classical machine learning model. Hyperparameter tuning was performed using RandomizedSearchCV. The tuned model achieved approximately 93.30% validation accuracy.

6. Final Test Result

Final Test Accuracy: 87% on the official 1,600-image test set.

Class	Precision	Recall	F1-Score
Glioma	0.93	0.66	0.77
Meningioma	0.82	0.85	0.84
No Tumor	0.84	1.00	0.91
Pituitary	0.91	0.97	0.94
Overall	0.88	0.87	0.87

7. Conclusion

The project demonstrates that traditional machine learning can classify brain MRI images using GLCM, intensity, and HOG features. Feature selection and Extra Trees tuning improved the model's performance, achieving a final 87% accuracy on the official test dataset. The model can also accept a single uploaded MRI image and predict one of the four classes.

Note: This project is for educational/research purposes and should not be used as a medical diagnostic system.